

THE INSTITUTE OF FINANCE MANAGEMENT



THE FACULTY OF COMPUTING AND MATHEMATICS

DEPARTMENT OF COMPUTER SCIENCE

BACHELOR OF COMPUTER SCIENCE

ACADEMIC YEAR 2025/2026

SUBJECT NAME: ADVANCED SYSTEM ANALYSIS AND DESIGN

SUBJECT CODE: CSU07211

GROUP ASSIGNMENT

STREAM MAIN

GROUP 3

SUB-GROUP 6

STUDENT'S NAME	REGISTRATION NUMBER
BRIGHTON JOHNSON MASSAWE	IMC/BCS/2520640
JERRY LUCAS MRUMAPILI	IMC/BCS/2520847
AISHA ZIRIWAN DAFFA	IMC/BCS/ 2511052
VICTOR DOMISIANI SENEDA	IMC/BCS/2523255
AGAPT DIONIZ SYLVANUS	IMC/BCS/2523974

QN. By using HTML, prepare User Interface prototype of your identified project idea.

As from the summary document of the project idea as required in the first question and being attached together with this document explaining and showing the functionality of the codes in respect to the output or display of them in the User interface. As for our project prototype html and CSS in html codes were used, it has two User interface;

- ✓ Login User Interface
- ✓ Student User interface

In which as in both of the interfaces, the features of the system project are clearly seen as follows;

➤ Working of the Login Interface

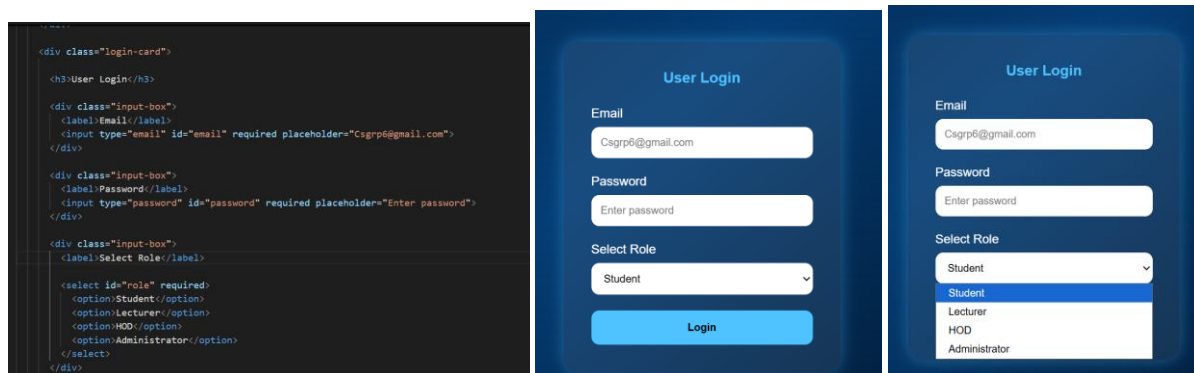
The login interface is the first page users interact with when accessing the UCS website. Functions of the Login Page to the users is as follows:

- Enter email address
- Enter password
- Access the student dashboard after login [When student option is selected]

How the Code Works

1. The HTML code creates: Input fields, Buttons, Containers and Labels.
2. CSS in html was used to: Add colors, Improve layout, Place style buttons and forms also make the interface attractive

Example of Login Inputs



Working Process

1. User opens the login page.
2. User enters email and password.
3. User clicks the Login button.
4. The system redirects the user to the student dashboard page.

➤ Working of the Student Dashboard/User Interface

After successful login, the student accesses the Student Dashboard interface. Functions of the Student Dashboard is where as the dashboard displays:

- Student profile information
- Navigation menu
- Announcements section
- Notifications
- Academic information sections

How the Dashboard Code Works

- 1.The HTML code is used to create: Sidebar, Header, Cards/Sections, Buttons and Student information panels.
2. The CSS code in html: Designs the dashboard layout, Adds background colors, Creates spacing and alignment also Improves visual appearance.

Example of Dashboard Section

The image displays the HTML code for a student dashboard on the left and the rendered dashboard on the right. The code is written in a dark-themed editor. The rendered dashboard is a dark blue interface for a student named Aisha Daffa. It features a header with a search bar, a notification icon, and a user profile. The main content area includes a welcome card with the student's name, a GPA of 4.1, and a list of today's classes: System Analysis & Design and Database Systems.

```
<main class="main-content">
  <header class="header-bar">
    <div class="search-container">
      <input type="text" placeholder="Search for messages, announcements, students, etc.">
    </div>
    <div class="header-actions">
      <div class="notification-icon">
        <span class="notification-badge"></span>
      </div>
      <div class="user-profile-header">
        <div class="user-avatar">AD</div>
        <div class="user-info">
          <span class="user-name">Aisha Daffa</span>
          <span class="user-role">Student</span>
        </div>
      </div>
    </div>
  </header>

  <main class="main-content">
    <div class="dashboard-grid">
      <div class="left-column">
        <section class="welcome-card">
          <div class="welcome-text">
            <h1>Hi Aisha </h1>
            <p>Welcome to UniSphere University Communication System. Stay updated with your classes, announcements, messages and important notifications.</p>
            <a href="#" class="view-announcements">View all announcements &arry</a>
          </div>
          <div class="gpa-circle">
            <span class="gpa-value">4.1</span>
            <span class="gpa-label">Current GPA</span>
          </div>
        </section>
        <section>
          <h2 class="section-title">Today's Classes</h2>
          <div class="list-container">
            <div class="list-item">
              <div class="item-details">
                <div class="item-icon-box"></div>
                <div class="item-meta">
                  <h4>System Analysis & Design</h4>
                  <p>08:00 - 10:00</p>
                </div>
              </div>
              <div class="item-meta">
                <p>Room: TH F<br/>Dr. Rose Tinabo</p>
              </div>
            </div>
            <div class="list-item">
              <div class="item-details">
                <div class="item-icon-box"></div>
                <div class="item-meta">
                  <h4>Database Systems</h4>
                  <p>11:00 - 13:00</p>
                </div>
              </div>
              <div class="item-meta">
                <p>Room: TH E<br/>Mr. Siphy Sommers</p>
              </div>
            </div>
          </div>
        </section>
      </div>
    </div>
  </main>
</div>
```

➤ Navigation Between Pages

The UCS website uses links to connect the login page and student dashboard and where as to ensure the user must first fill the credentials before jumping to other pages slight knowledge in javascript in html was used.

Example

```

<button class="login-btn"onclick="verifyCredentials()">Login</button>
</div>
<script>
function verifyCredentials() {
    event.preventDefault();

    var userEmail = document.getElementById("email").value;
    var userPassword = document.getElementById("password").value;
    var userRole = document.getElementById("role").value;

    if (userEmail === "Csgrp6@gmail.com" && userPassword === "123456" && userRole === "Student") {
        window.location.href = "Student-user-interface.html";
    } else {
        alert("Invalid email account, password match, or system role assignment!");
    }
}
</script>

```

Working

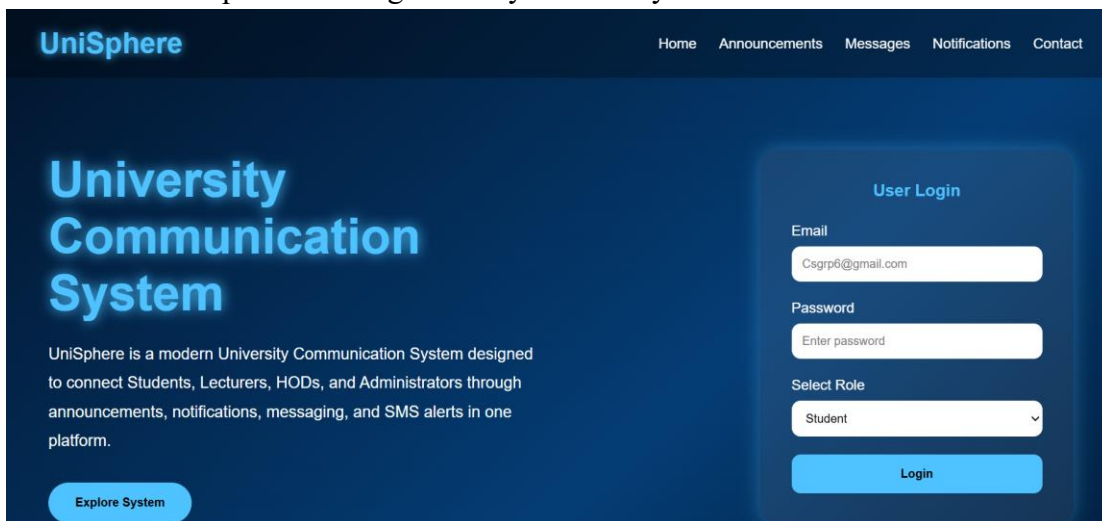
When the Login button is clicked, the system opens the Student Dashboard webpage. This allows smooth movement from one interface to another.

➤ User Interface Design

The UCS website was designed with:

- Modern layout
- Blue themed colors
- User-friendly structure
- Responsive arrangement of sections

The interface helps users navigate the system easily.



➤ Challenges Encountered

Some challenges faced during development include:

- Linking login page with dashboard
- Designing the dashboard layout
- Styling the interface using CSS
- Arranging sections properly

These problems were solved through testing and code adjustments.

➤ Conclusively

The UniSphere UCS website successfully demonstrates the development of a university web interface using HTML and CSS in html.

The system currently includes:

- Login Interface
- Student Dashboard/User Interface

The project helped improve skills in:

- Web design
- HTML coding
- CSS styling in html
- User interface development

Future improvements may include:

- Database connection
- Real authentication system
- Messaging features [SMS gateways]
- Notifications system
- Admin and lecturer dashboards

Whereby such features that make up the system are already identified as in the login user interface such as follows respectively to their codes;

```
</section>

<section class="Features">

  <h2>System Features</h2>

  <div class="feature-container">

    <div class="feature-box">
      <h3>Messaging</h3>
      <p>
        Send and receive messages between students,
        lecturers, and administrators.
      </p>
    </div>

    <div class="feature-box">
      <h3>Announcements</h3>
      <p>
        View important university announcements and updates instantly.
      </p>
    </div>

    <div class="feature-box">
      <h3>SMS Alerts</h3>
      <p>
```

```
    <h3>SMS Alerts</h3>
    <p>
      Receive emergency alerts and OTP verification messages through SMS.
    </p>
  </div>

  <div class="feature-box">
    <h3>Notifications</h3>
    <p>
      Get real-time notifications about schedules,
      events, and university activities.
    </p>
  </div>
</div>

</section>

<footer>
  <p>© 2026 UniSphere - University Communication System @ all rights reserved</p>
</footer>

</body>
</html>
```

Explore System

Login

System Features

Messaging

Send and receive messages between students, lecturers, and administrators.

Announcements

View important university announcements and updates instantly.

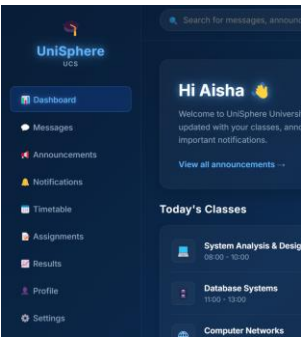
SMS Alerts

Receive emergency alerts and OTP verification messages through SMS.

Notifications

Get real-time notifications about schedules, events, and university activities.

```
<aside class="sidebar">
  <div class="logo-container">
    <div class="logo-icon">
    <div class="logo-title">UniSphere</div>
    <div class="logo-subtitle">UCS</div>
  </div>
  <ul class="nav-links">
    <li class="nav-item active"><a href="#">Dashboard</a></li>
    <li class="nav-item"><a href="#">Messages</a></li>
    <li class="nav-item"><a href="#">Announcements</a></li>
    <li class="nav-item"><a href="#">Notifications</a></li>
    <li class="nav-item"><a href="#">Timetable</a></li>
    <li class="nav-item"><a href="#">Assignments</a></li>
    <li class="nav-item"><a href="#">Results</a></li>
    <li class="nav-item"><a href="#">Profile</a></li>
    <li class="nav-item"><a href="#">Settings</a></li>
  </ul>
</aside>
```



```
<div class="todo-list">
  <div class="todo-item checked" style="border-left-color: var(--color-success);">
    <div class="todo-left">
      <div class="checkbox-round"></div>
      <div>
        <div class="todo-title">Complete Database Assignment</div>
        <div class="todo-code">DBMS CSU07208</div>
      </div>
    </div>
    <div class="todo-date">by 5 June 2026</div>
  </div>
  <div class="todo-item" style="border-left-color: var(--color-warning);">
    <div class="todo-left">
      <div class="checkbox-round"></div>
      <div>
        <div class="todo-title">Prepare for Networks Quiz</div>
        <div class="todo-code">CSU07209</div>
      </div>
    </div>
    <div class="todo-date">by Tomorrow</div>
  </div>
</div>
```

